



Báo cáo dự án môn Natural Language Processing

Image Captioning

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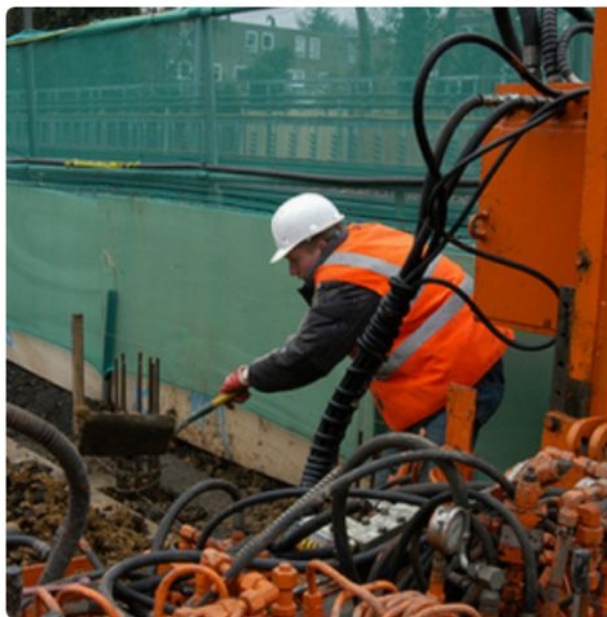
05

Demo

01 | Introduction



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."

Application:

- ❖ chú thích cho hình ảnh trên mạng xã hội,
- ❖ hỗ trợ cho người mù thông qua mô tả hình ảnh
- ❖ ứng dụng trong lĩnh vực y tế, nông nghiệp thông minh, và nhiều lĩnh vực khác

02 | Dataset

Flickr8k: 8000 ảnh

- **trainset: 6000 ảnh**
- **valset: 1000 ảnh**
- **testset: 1000 ảnh**

02 | Dataset

Flickr8k: 8000 ảnh

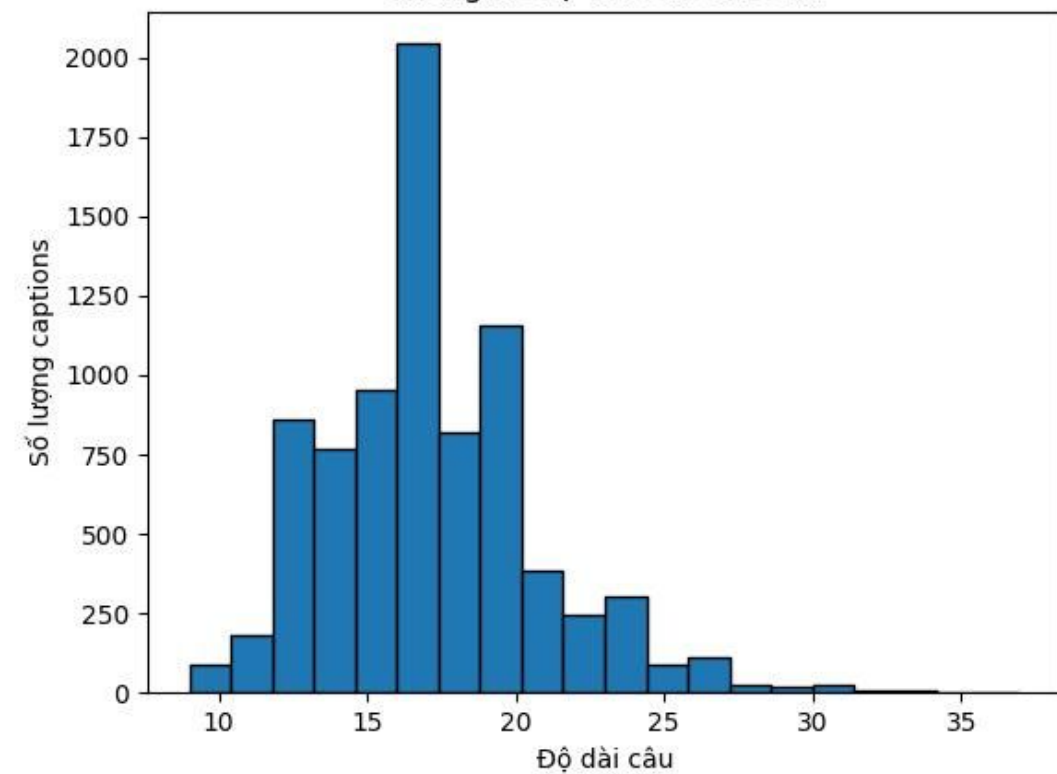
- **trainset: 6000 ảnh**
- **valset: 1000 ảnh**
- **testset: 1000 ảnh**

- ❖ a black and white border collie catches a frisbee in front of an audience
- ❖ a dog jumps to catch a frisbee while many people watch
- ❖ an agile dog catches a frisbee while a crowd of onlookers watches closely
- ❖ a brown and white dog catches a frisbee in it mouth in front of a group of people
- ❖ a brown and white dug jumping up to catch a frisbee while an audience watches



02 | Dataset

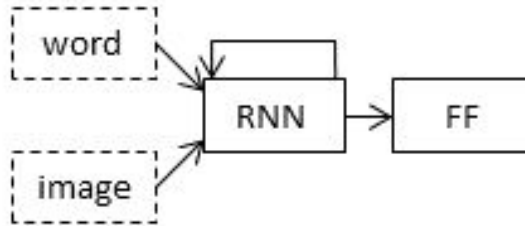
Thống kê độ dài của các câu



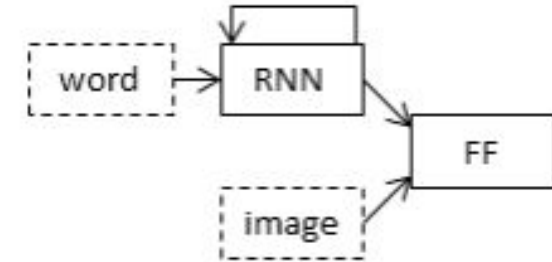
		Xi	Yi	
i	Image feature vector	Partial Caption	Target word	
1	Image_1	startseq	the	data points corresponding to image 1 and its caption
2	Image_1	startseq the	black	
3	Image_1	startseq the black	cat	
4	Image_1	startseq the black cat	sat	
5	Image_1	startseq the black cat sat	on	
6	Image_1	startseq the black cat sat on	grass	
7	Image_1	startseq the black cat sat on grass	endseq	
8	Image_2	startseq	the	data points corresponding to image 2 and its caption
9	Image_2	startseq the	white	
10	Image_2	startseq the white	cat	
11	Image_2	startseq the white cat	is	
12	Image_2	startseq the white cat is	walking	
13	Image_2	startseq the white cat is walking	on	
14	Image_2	startseq the white cat is walking on	road	
15	Image_2	startseq the white cat is walking on road	endseq	

03 | Method

Where to put the image in an image caption generator



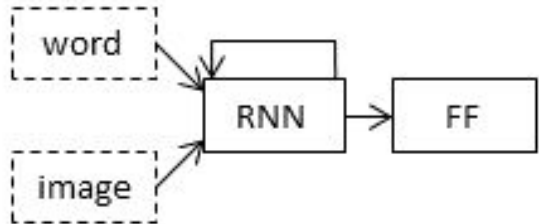
(a) Conditioning by injecting the image means injecting the image into the same RNN that processes the words.



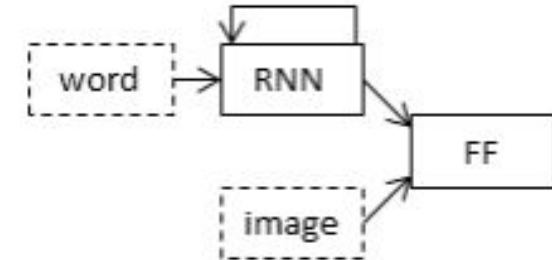
(b) Conditioning by merging the image means merging the image with the output of the RNN after processing the words.

03 | Method

Where to put the image in an image caption generator

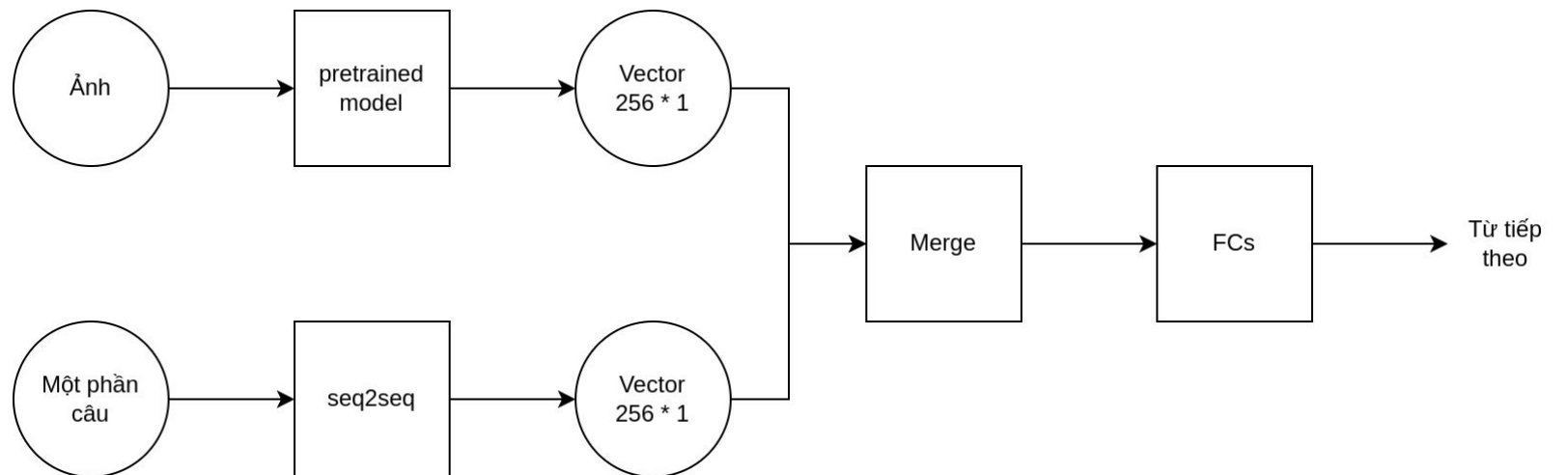


(a) Conditioning by injecting the image means injecting the image into the same RNN that processes the words.



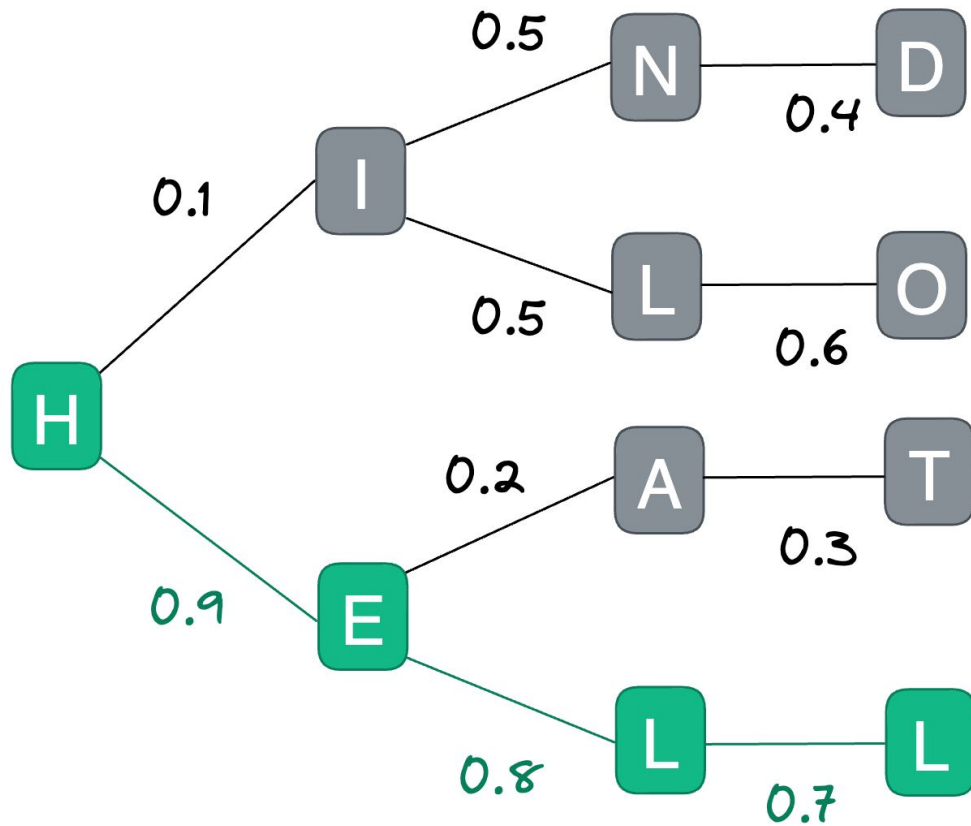
(b) Conditioning by merging the image means merging the image with the output of the RNN after processing the words.

Model



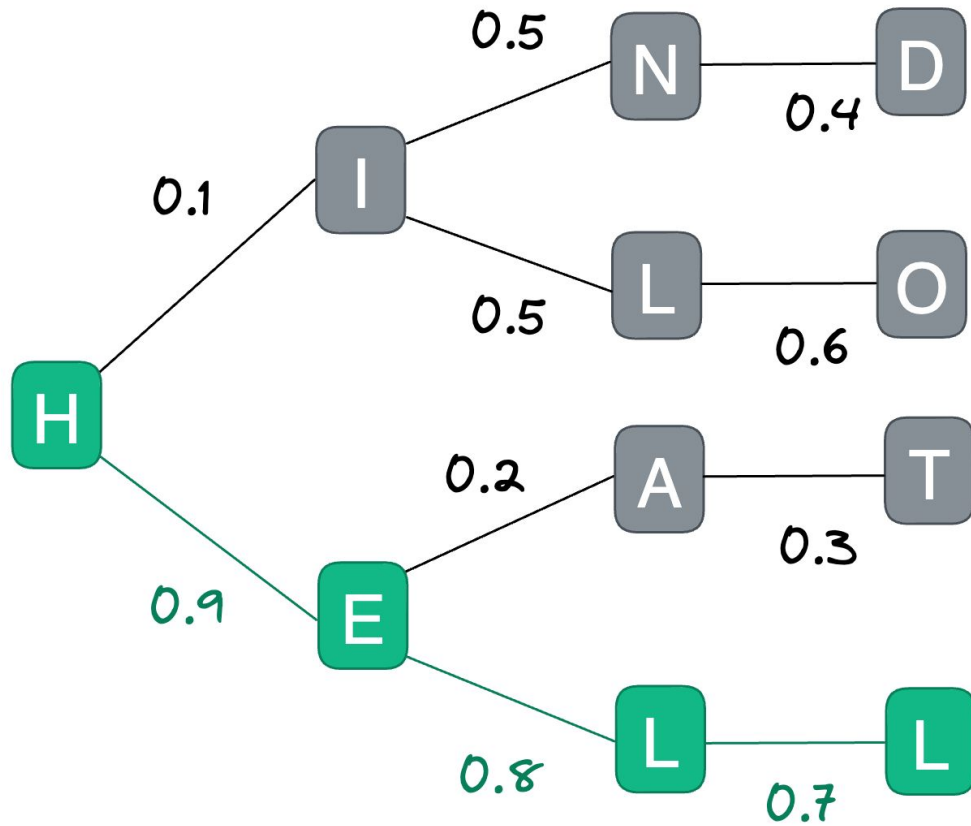
03 | Method: Search

Greedy Search

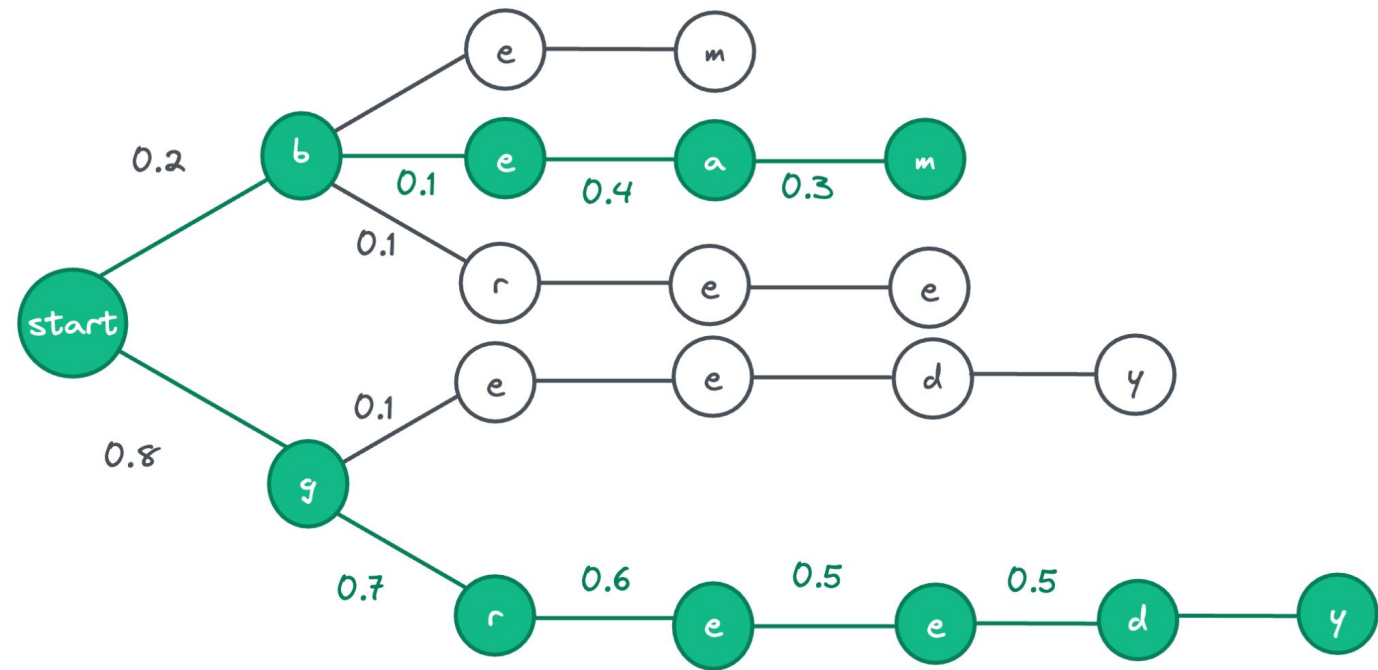


03 | Method: Search

Greedy Search

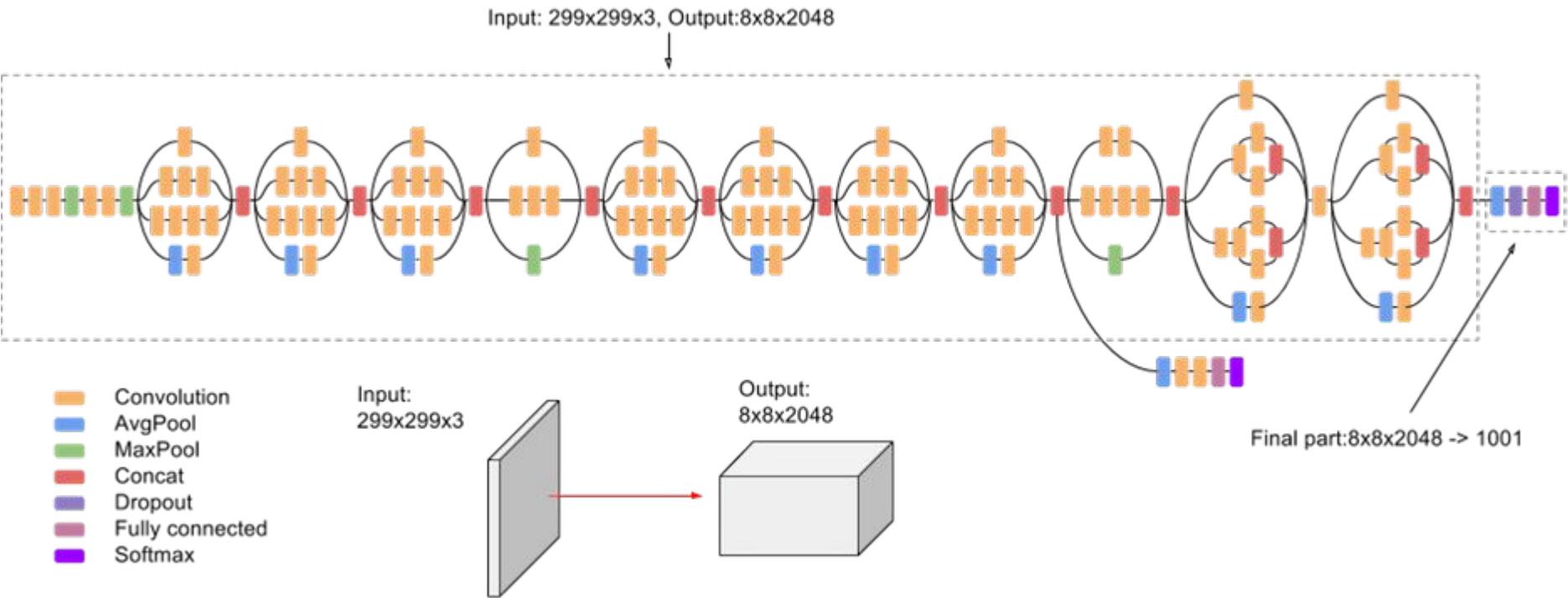


Beam Search



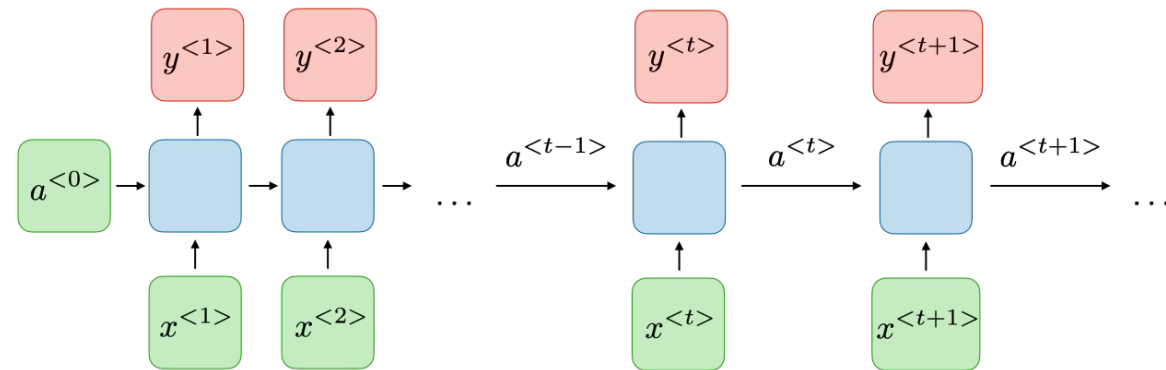
03 | Method: Image Embed

- ❖ Inception-V3
- ❖ Resnet

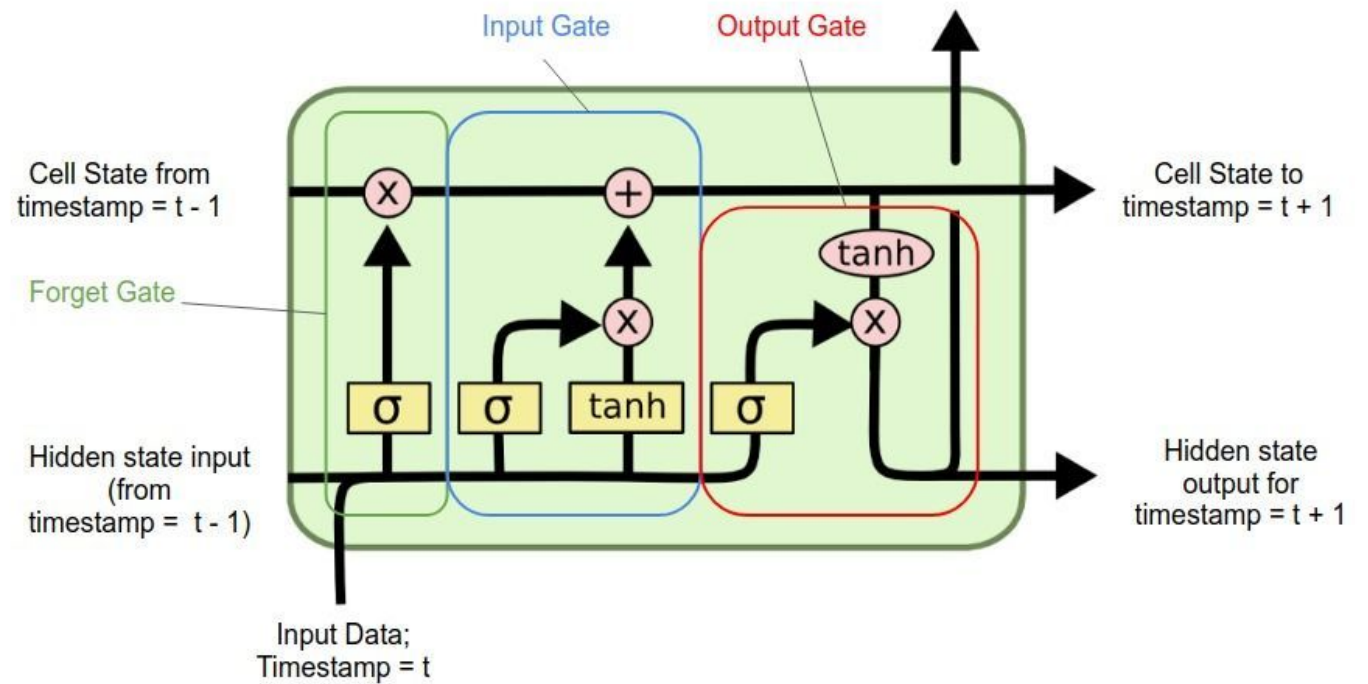


03 | Method: Seq2Seq

Recurrent neural network

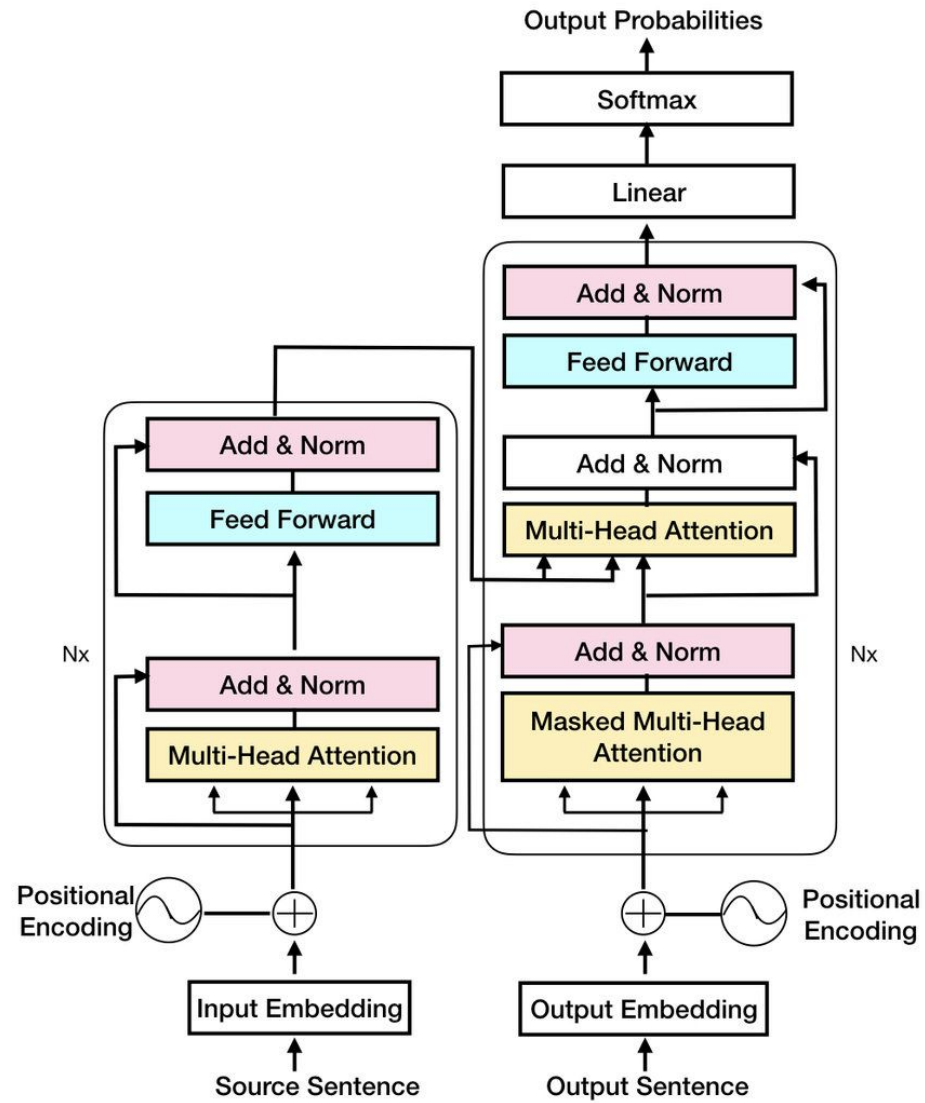


Long short term memory



03 | Method: Seq2Seq

Transformer



04 | Experiment

$$\text{BLEU} = \underbrace{\min \left(1, \exp \left(1 - \frac{\text{reference-length}}{\text{output-length}} \right) \right)}_{\text{brevity penalty}} \underbrace{\left(\prod_{i=1}^4 \text{precision}_i \right)^{1/4}}_{\text{n-gram overlap}}$$

with

$$\text{precision}_i = \frac{\sum_{\text{snt} \in \text{Cand-Corpus}} \sum_{i \in \text{snt}} \min(m_{\text{cand}}^i, m_{\text{ref}}^i)}{w_t^i = \sum_{\text{snt}' \in \text{Cand-Corpus}} \sum_{i' \in \text{snt}'} m_{\text{pred}}^{i'}}$$

BLEU Metrics

BLEU Score (percentage)	Interpretation
< 10	Almost useless
10 - 19	Hard to get the gist
20 - 29	The gist is clear, but has significant grammatical errors
30 - 40	Understandable to good translations
40 - 50	High quality translations
50 - 60	Very high quality, adequate, and fluent translations
> 60	Quality often better than human

04 | Experiment

Text Embed Model	use_glove	Merge Operation	BLEU-1	BLEU-2	BLEU-3	BLEU-4
RNN	true	add	0.55241	0.37043	0.23651	0.1536
RNN	true	concat	0.54297	0.35271	0.22136	0.13712
LSTM	true	add	0.57097	0.38688	0.25114	0.16337
LSTM	true	concat	0.56068	0.37466	0.23859	0.15176
TransformerEncoder	true	add	0.19013	0.10166	0.04107	0.0168
Transformer	false	transformer	0.48475	0.29834	0.17457	0.10973
Transformer	true	transformer	0.37894	0.20448	0.11482	0.07074

